Project Design Phase-I
Solution Architecture

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Team ID	699d5f7bfdf409fdbd1bebf5
Project Name	ResearchHub AI: Intelligent Research Paper Management and Analysis System using Agentic AI
Maximum Marks	4 Marks

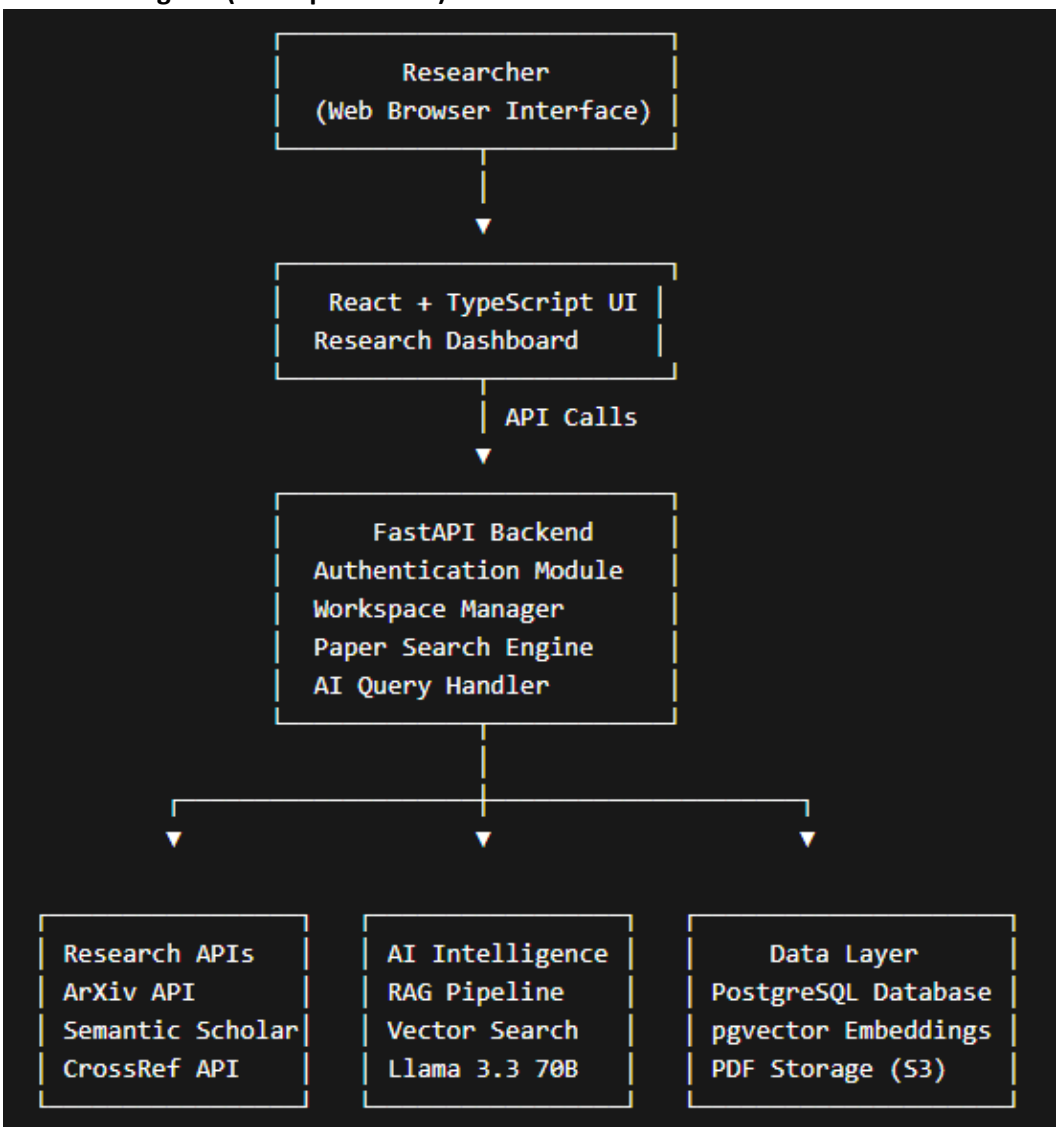
Solution Architecture:

Proposed Solution Architecture

The ResearchHub AI system follows a **multi-layer architecture** consisting of:

1. **Presentation Layer (User Interface)**
2. **Application Layer (Backend Services)**
3. **AI Intelligence Layer (RAG + LLM)**
4. **Data Layer (Databases and Storage)**
5. **External Integration Layer (Research APIs)**

Architecture Diagram (Conceptual Flow)



Architecture Components

1. User Interface Layer

The user interface provides a web-based platform where researchers can interact with the

system.

Key Functions

- User registration and login
- Research paper search
- Workspace management
- AI chatbot interaction
- Viewing research insights

Technologies

- React.js
- TypeScript
- Tailwind CSS

2. Application Layer

The backend layer manages all core application logic.

Key Functions

- User authentication and authorization
- Managing research workspaces
- Fetching research papers from APIs
- Processing AI queries
- Handling file uploads and metadata storage

Technologies

- FastAPI
- Python
- REST APIs

3. AI Intelligence Layer

This layer enables intelligent research analysis.

It uses a **Retrieval Augmented Generation (RAG)** architecture.

Components

- Document chunking
- Embedding generation
- Vector similarity search
- LLM-based response generation

Technologies

- Groq Llama 3.3 70B
- Embedding models
- pgvector vector search

4. Data Layer

The data layer stores all application and research data.

Stored Data

- User accounts
- Research workspaces
- Research paper metadata
- AI conversation history
- Document embeddings

Technologies

- PostgreSQL
- pgvector
- Cloud object storage (S3)

5. External Integration Layer

The system integrates with external academic data sources.

External APIs

- ArXiv API – research paper metadata
- Semantic Scholar API – citation and paper details
- CrossRef API – research metadata

These APIs allow the system to provide up-to-date research information.

Data Flow Overview

1. Researcher logs into the system.
2. User searches for research papers.
3. Backend queries external research APIs.
4. Paper metadata is retrieved and displayed.
5. Selected papers are stored in the workspace.
6. User asks AI questions about the papers.
7. System retrieves relevant document embeddings.
8. AI model generates contextual research insights.
9. Results are returned to the user interface.

Key Architectural Characteristics

Scalability

The architecture supports scalable research workloads using cloud-based services and modular AI components.

Security

Secure authentication and encrypted data storage ensure the protection of research data.

Reliability

The system uses managed cloud infrastructure and API-based services to ensure stable operation.

Performance

Vector databases and optimized AI pipelines enable fast retrieval of relevant research information.